

How Far Can General-Purpose LLMs Go in Tigrinya? A Lexicon-Grounded Evaluation with Trivial Baselines

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Abstract

We evaluate eight general-purpose large language models on three lexicon-grounded Tigrinya tasks — part-of-speech tagging, morphosyntactic labelling, and English→Tigrinya translation — using a 5,783-row annotated bilingual lexicon. Our central methodological finding is that such evaluations are easy to over-read: with a permissive matching rule, models appear to reach 80–89% accuracy, but the same outputs score 0–82% under a first-token rule, and a constant-answer baseline reaches 53.7% while an English-only tagger that never sees the Tigrinya reaches 60.4%. Measured against those baselines, three of eight models demonstrate above-trivial POS performance and two exceed the English-only baseline on morphosyntactic labelling, by under three points. On translation all eight models score at or below chrF 0.81, while NLLB-200-3.3B, which supports Tigrinya, reaches chrF 16.05 on identical items with an identical scorer. We argue that low-resource evaluations without trivial baselines are uninterpretable, and release the lexicon, code, model outputs, and job logs.

1 Introduction

Tigrinya is a Semitic language of Eritrea and northern Ethiopia with approximately 9–10 million speakers and very little machine-readable linguistic infrastructure. This scarcity makes it a natural test of whether general-purpose LLMs generalise beyond high-resource languages, and a difficult one to measure: without established benchmarks it is easy to produce numbers that look like competence.

This paper makes three contributions.

1. **A lexicon-grounded benchmark** of 5,783 English–Tigrinya rows (4,331 English headwords, 3,767 Tigrinya forms), each carrying a part-of-speech label, derived from a digitised bilingual dictionary (Union of International Democrats, 2015).
2. **An evaluation of eight LLMs** across POS tagging, morphosyntactic labelling and translation, reported under three matching rules and against two trivial baselines.
3. **A methodological result:** on a lexicon-scale task, the choice of matching rule moves reported accuracy by up to 26 points, and two baselines that use no Tigrinya at all reach 53.7% and 60.4%. Papers in this setting that report neither are not interpretable.

The third contribution is the reason for this paper. An earlier version of this work reported results computed with a permissive matching rule and no baselines, and drew an architectural conclusion that does not survive either correction. The authors have requested its retraction; Section 7 states the relationship between the two.

2 Related work

2.1 Tigrinya and Ge’ez-script NLP

Tigrinya remains severely underrepresented in NLP: a recent survey of over 50 studies from 2011 to 2025 finds a field whose progress is driven almost entirely by discrete resource-creation milestones rather than by modelling advances (Gaim and Park, 2026), a pattern echoed in broader surveys of Ethiopian-language NLP (Tonja et al., 2023). The resources that exist are small and task-specific. Öktem et al. (2020) built a domain-specific Tigrinya→English NMT system using transfer learning from other Ge’ez-script languages, reporting a 1.3 BLEU improvement over a neural baseline; Tela et al. (2020) showed that an English XLNet fine-tuned on 10k examples reaches 78.88 F1 on Tigrinya sentiment, outperforming mBERT by 7 points; TiGQA contributed 2,685 expert-annotated question–answer pairs from 537 context paragraphs (Teklehaymanot et al., 2024); and VEXMLM demonstrated that vocabulary extension with 30k Ge’ez-script subwords substantially improves XLM-R on Amharic and Tigrinya, attributing the baseline weakness to high out-of-vocabulary rates under Latin-centric tokenizers (Teklehaymanot et al., 2026b). Tigrinya also appears as one slice of multilingual benchmarks — MasakhaPOS, the largest POS dataset for African languages, covering 20 typologically diverse languages under UD guidelines (Dione et al., 2023), and Belebele, a parallel multiple-choice reading-comprehension set in 122 language variants (Bandarkar et al., 2024). Our lexicon complements these: it is not running text, and it tests the dictionary-headword level that sits beneath all of them. The structural reason such resources are scarce at all is the one Joshi et al. (2020) documented — representation in NLP venues tracks resource availability, calling into question the “language agnostic” status of current systems.

2.2 LLM evaluation for low-resource languages

NLLB-200 added Tigrinya (`tir_Ethi`, Ge’ez script, Afro-Asiatic/Semitic) to a 200-language translation model (NLLB Team et al., 2022), evaluated on FLORES, whose professionally translated, multilingually aligned design set the standard for low-resource MT benchmarking (Goyal et al., 2022). Against that backdrop, several studies have established that general-purpose LLMs lag purpose-built MT on exactly the languages we study. Robinson et al. (2023) evaluated ChatGPT on 204 FLORES-200 varieties and found NLLB superior on 169 of 201 languages (84.1%), by an average of 11.9 chrF, with the gap widest for African languages and the Ge’ez script specifically; they further report that five-shot prompting improved on zero-shot by only 0.88 chrF on average across 203 directions, which is consistent with our own few-shot probe moving gemma-7b from chrF 0.16 to 0.66 with exact match unchanged. Hendy et al. (2023) reach the same conclusion for GPT models across eighteen directions — competitive for high-resource, limited for low-resource — and Zhu et al. (2023) find GPT-4 beating NLLB in 40.91% of directions overall while still trailing badly on the long tail. Adebara et al. (2025) extend this to a continent-scale African benchmark and attribute the residual disparities to data inequities rather than model scale. Our Section 5.3 result — three NLLB checkpoints at chrF 14.1–16.1 against eight LLMs at ≤ 0.81 — is this finding at lexicon granularity, where the LLM floor is not merely lower but absolute.

A separate strand asks whether LLM evaluations in these settings are conducted carefully enough. LArABench benchmarked 33 Arabic tasks across 61 datasets and explicitly computed a random baseline “to determine if the LLMs predictions are not merely the result of chance” (Abdelali et al., 2023). We regard that step as mandatory rather than optional, and Section 5 shows why.

2.3 Trivial baselines and spurious cues

The methodological core of this paper has a direct precedent in natural language inference. Poliak et al. (2018) showed that a hypothesis-only model — one that never sees the premise, a degenerate solution by construction — significantly outperforms a majority-class baseline on a number of NLI datasets; Gururangan et al. (2018) showed the same artefacts let a simple text classifier label the hypothesis alone correctly in about 67% of SNLI and 53% of MultiNLI, concluding that model success “has been overestimated”. McCoy et al. (2019) demonstrated that strong test-set scores can rest on fallible syntactic heuristics that collapse on controlled sets, and Niven and Kao (2019) found BERT’s 77% on argument reasoning comprehension to be entirely accounted for by spurious statistical cues. Our English-only tagger is a hypothesis-only baseline in this sense: the prompt format `Phrase: {Tigrinya} ({English})` leaves the part of speech largely recoverable from the gloss, and the baseline reaching 60.38 is the lexicon-task analogue of the 67% SNLI result.

For POS tagging specifically, Kann et al. (2020) make the closest argument to ours: weakly supervised taggers reported to perform “almost as well as supervised ones” were evaluated on languages unlike truly low-resource ones, and on 15 genuinely low-resource languages the best model gets fewer than half the words right, with a strongest baseline macro-average of 39.11%. They also note that such methods presuppose “high-coverage and almost error-free dictionaries” — precisely the assumption our Section 3.3 annotator notes undermine, where the source dictionary contains mislabelled parts of speech and outright mistranslations. At the level of benchmark design, Reuel et al.’s assessment of 46 best practices across 24 benchmarks identifies stating random performance as a scored criterion, since without it one cannot separate capability from chance or metric design (Reuel et al., 2024).

2.4 Scoring rules, answer extraction and prompt artefacts

Our finding that the matching rule moves reported accuracy by up to 26 points on fixed outputs belongs to a growing literature on evaluation mechanics. Tam et al. (2024) show that constraining LLMs to structured output formats significantly degrades measured reasoning, and disentangle format errors from content by using an LLM as a “perfect parser” rather than a regex — the mirror image of our problem, where permissive parsing manufactures competence instead of destroying it. Kirouane and Petrocheilos (2026) document the same failure mode concretely in a low-resource setting: a scorer that scanned the whole response instead of the requested answer line produced a +29.8 pp artefact, one-directional against verbose outputs, and they conclude that “any benchmark that requests an answer format must score that format first and report how often it was absent”. Our first-token rule is that recommendation applied; our exact / first-token / set-intersection triple reports the spread it would otherwise hide.

The prompt-echo column in Section 5.2 relates to known label-copying behaviour. Zhao et al. (2021) show that few-shot LLM accuracy is destabilised by bias towards answers appearing in the prompt or common in pretraining, varying “from near chance to near state-of-the-art” with prompt format alone. When the gold distribution is 53.7% `noun` and the prompt example contains the word `noun`, a model that copies the example is scored correct by construction under set-intersection — gemma-7b does this on 97.4% of answers. We therefore treat echo rate as a reportable diagnostic rather than a footnote.

2.5 Metrics and agreement statistics

We report chrF (Popović, 2015) because character n -gram F-score is tokenisation-independent and therefore appropriate for Ge’ez-script targets averaging 1.37 whitespace tokens, where word-level BLEU (Papineni et al., 2002) is near-undefined; we follow Post’s argument that BLEU is a parameterised family whose configurations are not comparable across papers unless

the scheme is pinned down (Post, 2018), and report character-level BLEU alongside chrF for this reason. Agreement is reported as Cohen’s κ (Cohen, 1960). We note the standard caveat that κ is hard to interpret under class imbalance — the “kappa paradox” — and that absolute thresholds are not meaningful without a contextual reference (Wong et al., 2021); this is why Section 3.3 reports κ beside raw observed agreement and beside the filled-label-only variant, rather than reporting a single coefficient.

2.6 Dictionaries as evaluation resources

Digitised bilingual dictionaries are a standard bottom-up resource for languages lacking parallel corpora: Wickramasinghe and de Silva (2023) argue that for low-resource pairs it is “more feasible to move in the bottom-up direction where finer granular pairs such as dictionary datasets are developed first”, and Alnajjar et al.’s Ve’rdd addresses the complementary problem of grassroots paper dictionaries edited by many hands, which require re-evaluation rather than direct ingestion (Alnajjar et al., 2020). Our source exhibits exactly the defects that motivate that work — a direction-swapped second block, inline gender markers embedded in the target field, and annotator-flagged mistranslations — and Sections 3.1–3.3 document them rather than silently cleaning them.

2.7 Relation to the earlier version

An earlier version of this work, *LLM Probe: Evaluating LLMs for Low-Resource Languages* (Teklehaymanot et al., 2026a), introduced the lexicon and the four-task framework used here. It reported POS accuracies of 73.0–80.0 and morphosyntactic accuracies of 70.5–77.0 under a permissive matching rule with no baselines, assigned different model subsets to different tasks, and concluded that sequence-to-sequence models “excel in morphosyntactic analysis and translation quality”. Section 7 states which of those claims survive re-scoring and which do not.

3 Dataset

3.1 Source and construction

The lexicon was digitised from the Swansea Tigrinya–English dictionary (Union of International Democrats, 2015), with additional entries contributed by native-speaker linguists, and carries the source dictionary’s own part-of-speech abbreviations. The digitised file, `Combined_POS_Lexicon.csv`, holds 7,234 rows in two blocks: 3,587 English→Tigrinya entries followed by 3,647 Tigrinya→English entries. The two blocks share column headers, so in the second block the column named `English` holds Ge’ez script and the column named `Tigrigna` holds English.

Provenance. That CSV is a flattening of `POS_english_to_tigrigna_Annotated.xlsx`, the working annotation workbook, whose eleven part-of-speech sheets hold the same 7,234 rows with the same per-category counts (3,929 noun, 1,784 verb, 993 adjective, 194 adverb, ...). The flattening concatenates those sheets in workbook order and lifts the inline gender/number markers that the workbook keeps inside the Tigrinya field — (nf) ወኅሰት, ከሳሰት — out into the `POS_CATEGORIES` column, which is where the finer labels (nm) noun masculine, (nf) noun feminine and (npl) noun plural come from. The two files differ on exactly the 247 rows carrying such a marker. The column swap described above is present in the workbook as well: 3,635 of its 7,234 rows (50.2%) carry Ge’ez script in the `English` column, so the defect originates in digitisation, not in the CSV export.

`scripts/build_merged_eval_set.py` detects the second block by script, reorients it, and merges both directions, writing **5,783 rows** to `data/gold_labels/`. Expanding comma-

separated senses gives 5,864 atomic pairs across 4,331 English headwords and 3,767 Tigrinya forms.

7,234 counts rows, not independent annotations. Of the 3,646 unique reverse-direction pairs, 1,448 (39.7%) also occur in the forward block.

Three counts appear in this paper, and they are not interchangeable. The gold files hold **5,783 rows**; `--verify` confirms the script reproduces them exactly. Deduplication there keys on (English, Tigrinya, features) rather than the pair alone, so distinct senses of one form survive as separate items: *estimate* → ግምት is kept as both noun and verb, and *grocer* → በዓል ድህን as both noun and noun, masculine.

Each task script then applies its own pair-level deduplication at load time, dropping 7 duplicate (English, Tigrinya) pairs and 1 row with no English side, leaving **5,775 evaluated items** for translation and lexical alignment. POS tagging and morphosyntactic labelling were run against a further reduction in which a multi-sense entry is represented by its first sense — (*a little*, ንእሽቶይ, ቁሩብ, ውሕድ) becomes (*a little*, ንእሽቶይ) — giving **5,701 items**.

That sense-split file was an intermediate and was not originally committed, so the task scripts as they stand would evaluate 5,775 rather than the 5,701 behind Sections 5.1 and 5.2. `scripts/build_sense_split_eval_set.py` reconstructs it from the committed gold and verifies the result row-for-row against the saved model outputs: 5,701 items, no row missing, none extra, no label mismatch.

3.2 What the annotations do and do not contain

The POS labels are the source dictionary’s, not independently re-annotated by us. They are coarse: **noun** accounts for 3,109 of 5,783 labelled entries (53.7%), **verb** 1,542, **adjective** 879.

Morphosyntactic features are **sparse**. Only 185 entries carry gender, number or plurality; the remaining 5,598 (96.8%) carry a value identical to the POS label. We therefore do not treat morphosyntactic labelling as a distinct linguistic dimension, and report it in Section 5.2 as what it measures: a second POS task with a different prompt.

3.3 Inter-annotator agreement

A 500-item subset of the lexicon was double-annotated for part of speech, gender, number, category agreement and lexical alignment. The records are released as `POS_english_to_tigrigna_Annotated.xlsx` — two annotator sheets, an adjudicated sheet and the annotation guide that defines each value — and `scripts/compute_iaa.py` recomputes every figure below from them.

An annotator sometimes left a cell empty where the other assigned a label. Those are not matching judgements, so we count them as disagreements and report Cohen’s κ over all 500 items, with a blank treated as its own category (Table 1).

Restricting instead to the items both annotators labelled raises every dimension — 0.9076, 1.0000, 1.0000, 0.9815, 0.9850 on 500, 420, 428, 445 and 474 items. We report the stricter figures because the difference between the two columns is largely a difference in what each annotator left blank: Annotator 1 omits 32 gender, 32 number and 46 agreement labels that Annotator 2 supplies, against 6, 4 and 3 in the other direction. Discarding those rows would hide the main source of divergence.

Three properties of this subset bound what the coefficients support, and we state them rather than leave them to be discovered.

Dimension	κ	Observed
Part of speech	0.9076	0.9500
Gender	0.8079	0.9240
Number	0.6762	0.9280
Category agreement	0.5912	0.9000
Lexical alignment	0.8487	0.9480

Table 1: Inter-annotator agreement on the 500-item subset, blanks counted as disagreements. All five dimensions are computed over all 500 items.

Label	Checkpoint	Arch.	Inst.
gemma-2b	google/gemma-2b-it (Gemma Team, Google DeepMind, 2024)	causal	yes
gemma-7b	google/gemma-7b-it (Gemma Team, Google DeepMind, 2024)	causal	yes
mistral-7b	Mistral-7B-Instruct-v0.2 (Jiang et al., 2023)	causal	yes
qwen-7b	Qwen1.5-7B-Chat (Bai et al., 2023)	causal	yes
falcon-7b	falcon-7b-instruct (Almazrouei et al., 2023)	causal	yes
mt5-small	google/mt5-small (Xue et al., 2021)	seq2seq	no
mt5-large	google/mt5-large (Xue et al., 2021)	seq2seq	no
byt5	google/byt5-small (Xue et al., 2022)	seq2seq	no

Table 2: Evaluated models. Architecture and instruction-tuning status are perfectly confounded in this set.

The two columns are not independent of the adjudication. On the 25 part-of-speech items where the annotators differ, the adjudicated sheet reproduces Annotator 2 on all 25 and Annotator 1 on none; on the three gender, agreement and alignment disagreements it reproduces Annotator 1. The adjudication is assembled from the two columns rather than decided separately, so it cannot serve as an independent third judgement.

Agreement on filled labels is near-total. Where both annotators assigned a gender or number value, they never differ — 0 disagreements across 420 and 428 items. The divergence between them is entirely in coverage, not in choice of label, which is why κ falls as far as it does once blanks count.

The subset is not a random sample. Tracing each item to its source sheet shows the 500 are the alphabetical head of each category: all 9 prepositions, all 10 pronouns and all 6 interjections, then 305 adjectives and 150 nouns from the tops of those sheets. The corpus is 54% noun and 25% verb; this subset is 61% adjective and contains **no verbs**. The coefficients therefore describe agreement on this subset and are not an estimate for the corpus. A stratified re-annotation covering verbs would be required for that, and we report it as future work rather than extrapolating (Section 9).

The workbook also carries the annotators’ free-text notes on 201 of the 500 items, documenting mislabelled parts of speech in the source dictionary, inflected forms given as citation forms, and outright mistranslations (*at least* $\rightarrow$ at least , which means *at most*). These support the caveat in Section 3.2 that the POS labels are the source dictionary’s and were not independently verified.

4 Experimental setup

Eight models, all loaded through the Hugging Face **transformers** interface (Wolf et al., 2020) on an Ampere-class GPU node (4×A100, 32 CPU cores, 128 GB RAM) under SLURM (Table 2).

We do not compare architectures. All five causal models are instruction-tuned and all three sequence-to-sequence models are raw pretrained checkpoints; the two factors are perfectly confounded in this model set and no result here can separate them. The three base seq2seq models receive span-infilling prompts matched to their pretraining objective; they nonetheless produce almost no usable output, which we report as a property of those checkpoints on this task rather than of their architecture.

Generation is greedy with `max_new_tokens=128`, `min_new_tokens=1`. `xlm-roberta-base` (Conneau et al., 2020) was included initially but fails on all tasks — its prompts contain no `<mask>` token — and is excluded.

4.1 Scoring

Model answers are frequently verbose: gemma-7b’s median POS answer is 14 words. We report three rules.

- **exact** — the cleaned answer equals the gold label.
- **first-token** — the model’s first word is a gold label word. This is the answer the model commits to, and is our headline rule.
- **set-intersection** — any word of the answer is a gold label word. This is permissive: an answer naming several parts of speech is correct as soon as one fits. We report it only to show the gap.

4.2 Baselines

Two reference points, both of which a model must beat to demonstrate Tigrinya knowledge:

- **Majority** — always answer the most frequent gold label (**noun**).
- **English-only** — an NLTK (Loper and Bird, 2002) English POS tagger applied to the English gloss alone. The prompt format is **Phrase: {Tigrinya} ({English})**, and a dictionary headword’s part of speech is largely recoverable from the English word, so this baseline uses no Tigrinya whatsoever.

5 Results

5.1 POS tagging ($n = 5,701$)

Three models exceed both baselines: mistral-7b, gemma-7b, gemma-2b. Falcon-7b and qwen-7b fall below the majority baseline; all three seq2seq checkpoints are at or near zero.

The exact column shows how brittle these numbers are. Gemma-7B scores 0.00% exact and 79.39% first-token: it always answers in a sentence, never a bare label. A reader given only one of these columns would draw opposite conclusions.

5.2 Morphosyntactic labelling ($n = 5,701$)

Because the gold is 96.8% identical to the POS label (Section 3.2), this is effectively a second POS task.

Only two models exceed the English-only baseline of 61.57, by 3.02 and 1.75 points; the remaining six fall below it, and five fall below the majority baseline of 52.38.

The final column measures how often an answer contains the prompt’s own example, **preposition, noun, singular**. Gemma-7b echoes it in 97.4% of answers; because the example contains **noun**, and **noun** is the majority gold label, set-intersection scores those echoes as correct.

Model	exact	first-tok	set-int
mistral-7b	33.71	81.56	85.91
gemma-7b	0.00	79.39	82.44
gemma-2b	71.71	72.86	73.29
falcon-7b	26.94	40.64	44.45
qwen-7b	17.00	17.00	17.07
mt5-large	0.00	0.09	1.75
byt5	0.00	0.00	0.05
mt5-small	0.00	0.00	0.00
Majority (noun)		53.73	
English-only tagger		60.38	

Table 3: POS tagging accuracy (%) under three matching rules, with both trivial baselines.

Model	first-tok	set-int	echo
gemma-2b	64.59	78.02	1.2%
gemma-7b	63.32	89.42	97.4%
mistral-7b	54.92	81.09	3.7%
falcon-7b	39.17	76.28	19.5%
qwen-7b	34.33	54.45	0.4%
mt5-small	0.86	2.49	0.0%
mt5-large	0.11	0.74	0.0%
byt5	0.00	38.55	69.7%
Majority	52.38		
English-only tagger	61.57		

Table 4: Morphosyntactic labelling accuracy (%). The final column is how often an answer contains the prompt’s own example.

Its apparent 89.42 is largely an artefact of copying the prompt. ByT5’s 38.55 is the same artefact at 69.7% echo, and drops to 0.00 under first-token.

5.3 Translation, English→Tigrinya ($n = 5,775$)

Tigrinya references average 1.37 whitespace tokens, so word-level BLEU-4 is near-undefined; we report chrF and character-level BLEU (Table 5).

All eight general-purpose models are at the floor. NLLB-200, for which Tigrinya (`tir_Ethi`) is a supported target language, reaches chrF 16.05 on identical items with an identical scorer — roughly twenty times the best LLM.

Note that word-level BLEU-4 is 0.24 for NLLB-3.3B, which translates 16.24% of items exactly. **Word-level BLEU is not usable at lexicon scale**, and a paper reporting it would show a working system as scoring zero.

We verified the LLM failure is not one of output format. Few-shot prompting with eight exemplars and an explicit format constraint moved gemma-7b from chrF 0.16 to 0.66 with exact match unchanged at 0.00; extracting the Ge’ez span from verbose answers recovered the reference in 2 of 5,775 items. The median character-level similarity between model output and reference is 0.000: the models emit Ge’ez-shaped strings that are not the target words (ገህል for ብዙሐ, ልልል as repetition).

Model	chrF	BLEU _c	BLEU _w	exact
NLLB-3.3B	16.05	14.23	0.24	16.24
NLLB-1.3B	15.01	13.29	0.23	15.00
NLLB-600M	14.11	12.59	0.25	14.42
byt5	0.81	0.01	0.00	0.05
gemma-2b	0.39	0.01	0.01	0.07
mistral-7b	0.26	0.00	0.00	0.03
mt5-small	0.20	0.04	0.02	0.05
gemma-7b	0.16	0.00	0.00	0.03
mt5-large	0.13	0.04	0.01	0.02
falcon-7b	0.13	0.00	0.00	0.03
qwen-7b	0.04	0.00	0.00	0.28

Table 5: Translation quality. All values on a 0–100 scale.

5.4 Lexical alignment

We report this task for completeness but do not draw conclusions from it: 2,208 of 5,783 alignment labels (38%) are positional fallbacks (**AlignmentSource: auto**) rather than annotation. Format accuracy ranges from 0.0383 (mt5-large) to 0.5484 (gemma-2b); semantic accuracy from 0.0012 (mt5-small) to 0.8876 (mistral-7b). The divergence between the two for individual models is large enough that we regard the task as not yet well-posed.

6 Discussion

The matching rule is a reporting decision, not an implementation detail. On fixed model outputs, moving from set-intersection to first-token costs gemma-7b 3.05 points on POS and 26.10 on morphosyntax; moving to exact match costs it everything. A paper that reports one number without naming the rule has not reported a result.

Trivial baselines are not optional. Two baselines using no Tigrinya at all — a constant answer and an English tagger — reach 53.73 and 60.38 on POS and 52.38 and 61.57 on morphosyntax. Five of eight models fall below the stronger baseline on POS and six of eight on morphosyntax. Without those reference points, the same table reads as evidence of cross-lingual competence.

Prompt echo is a measurable confound. Gemma-7b reproduces the prompt’s example in 97.4% of its morphosyntax answers, and because the example contains the majority gold label, a permissive scorer rewards it. We recommend reporting echo rate wherever a prompt contains a worked example.

Purpose-built MT still dominates at the lexicon level. NLLB-200 checkpoints reach chrF 14.11–16.05 where the best general-purpose model reaches 0.81 — more than an order of magnitude on chrF. For low-resource languages, model selection dominates model size.

What we can claim about Tigrinya competence. Three of eight models show above-baseline POS performance. Two exceed the English-only baseline on morphosyntactic labelling, by 3.02 and 1.75 points — a margin small enough that we would not rest a claim on it without a held-out re-run. None can translate into Tigrinya. Even NLLB reaches only 16.24% exact on a dictionary task, which indicates the difficulty is real rather than an artefact of our setup.

Table	Command (in scripts/)	Artifact	Job log
§3.1 counts	build_merged_eval_set.py --dry-run --verify	Combined_POS_ Lexicon.csv	—
§3.1 5,701 set	build_sense_split_eval_set.py --verify	*_sense_ split.json	—
§3.3 agreement	compute_iaa.py	*_Annotated.xlsx	—
§5.1, §5.2	rescore_tasks.py --models \$ROSTER	rescored_ 2026-09-28.json	eval_ 75167.out
§5.3 LLMs	compute_bleu.py	translation_ fidelity_bleu.json	eval_tf_ rerun_76683.out
§5.3 NLLB	probe_nllb_translation.py	nllb_probe_*.json	nllb_ sweep_77617.out
§5.3 few-shot	probe_fewshot_translation.py	fewshot_probe_ gemma-7b.json	fewshot_ probe_77601.out

Table 6: Each reported table, the command that regenerates it, the artifact it reads, and the SLURM job that produced the underlying generations.

7 Relation to an earlier version of this work

An earlier version of this work was published as *LLM Probe: Evaluating LLMs for Low-Resource Languages* at LLMs4SSH @ LREC 2026, pp. 224–234 (2026.llms4ssh-1.24) (Teklehaymanot et al., 2026a). Its POS tagging and morphosyntactic results were computed with a permissive matching rule, under which an answer counts as correct if any of its words matches a gold label word, and were reported without baselines. Re-scored under a first-token rule and measured against a constant-answer baseline and an English-only baseline, the central claims of that paper do not hold: only three of the eight systems exceed both baselines on part-of-speech tagging and two on morphosyntactic labelling, by a few points in each case, where the published version reported accuracies of 80–89% against no reference point at all. The architectural conclusion it drew is confounded with instruction-tuning, since the sequence-to-sequence checkpoints evaluated were never instruction-tuned. The authors have requested that it be retracted.

This paper reports the same model outputs, re-scored. It shares the lexicon and the saved generations with that version and supersedes it; no figure here is taken from it, and the dataset counts, matching rules and baselines are all regenerable from the artifacts listed in Section 8. Readers comparing the two should treat the earlier numbers as withdrawn rather than as an alternative measurement.

8 Reproducibility

Every number in this paper comes from a committed artifact. `$ROSTER` below is the evaluated roster, `gemma-2b,gemma-7b,mistral-7b,falcon-7b,qwen-7b,mt5-small,mt5-large,byt5`; passing it pins the tables to these eight models, since later runs add result files to the same directories.

Raw model outputs for every item are in
`results/evaluation_reports/`
`<task>/<task>_<model>.json`.

9 Limitations

- POS labels are the source dictionary’s, not independently re-annotated. The agreement coefficients in Section 3.3 describe a 500-item subset that is alphabetical and contains no verbs, and whose adjudication is assembled from the two annotator columns rather than

decided separately. They are not an estimate for the corpus; a stratified re-annotation covering verbs would be needed for that.

- Morphosyntactic features cover 185 of 5,783 gold rows.
- Lexical alignment gold is 38% positional fallback; Section 5.4 is not a result.
- Entries are dictionary headwords, not running text; performance here does not predict sentence-level performance.
- The model set cannot separate architecture from instruction-tuning. An instruction-tuned seq2seq model (`flan-t5`, `mt0`) would be needed.
- The English-only baseline shows the POS task is substantially solvable without Tigrinya. A harder task would withhold the English gloss.

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